

Project Development Phase

Preprocessing and Business Questions with Visualizations

Date	15 July 2026
Team ID	
Project Name	Plugging into the Future: An Exploration of Electricity Consumption Patterns
Maximum Marks	4 Marks

Preprocessing Steps

Step 1 – Connecting Dataset

Description

The electricity consumption dataset was imported into Tableau Public using the Text File connection. Tableau automatically recognized the dataset structure and loaded all records successfully.

The screenshot displays the Tableau Public interface for a data source named 'Consumption (tabDB)'. The sidebar on the left shows the 'Connections' section with 'localhost' selected, and the 'Database' section with 'tabDB' selected. The 'Table' section lists 'Consumption' as the selected table. The central pane shows the 'Fields' section with a list of fields: States, Regions, Latitude, Longitude, Dates, and Usage. The data preview table on the right shows the first 10 rows of data, including columns for States, Regions, Latitude, Longitude, Dates, and Usage. The 'Consumption (tabDB)' title is highlighted in the top right corner. The 'Fields' pane is highlighted in the bottom left corner. The 'Connections' section is highlighted in the top left corner.

States	Regions	Latitude	Longitude	Dates	Usage
Punjab	NR	31.5200	75.9800	01/02/2019	119.9
Haryana	NR	28.4500	77.0200	01/02/2019	130.3
Rajasthan	NR	26.4500	74.6400	01/02/2019	234.1
Delhi	NR	28.6700	77.2300	01/02/2019	85.8
UP	NR	27.6000	78.0500	01/02/2019	313.9
Uttarakhand	NR	30.3204	78.0500	01/02/2019	40.7
HP	NR	31.1000	77.1666	01/02/2019	30.0
J&K	NR	33.4500	76.2400	01/02/2019	52.5
Chandigarh	NR	30.7200	76.7800	01/02/2019	5.0
Chhattisgarh	WR	22.0904	82.1600	01/02/2019	78.7

Step 2 – Data Validation

Description

The imported dataset was verified by checking field names, data types, and record consistency. Date fields, state names, regions, and usage values were validated to ensure accurate visualization.

Consumption (tabDB)

Connection: ☐ Live ☒ Extract | Edit Refresh
Extract contains all data. 7/3/2026 3:34:09 PM

Filters: 0 | Add

Consumption 6 fields 16599 rows 10000 rows

Name: Consumption
Description: No description available.

Fields:

Type	Field Name	Physical Ta...	Rem...
+	States	Consumption	States
Abc	Regions	Consumption	Regions
+	Latitude	Consumption	latitude
+	Longitude	Consumption	longit...
+	Dates	Consumption	Dates
#	Usage	Consumption	Usage

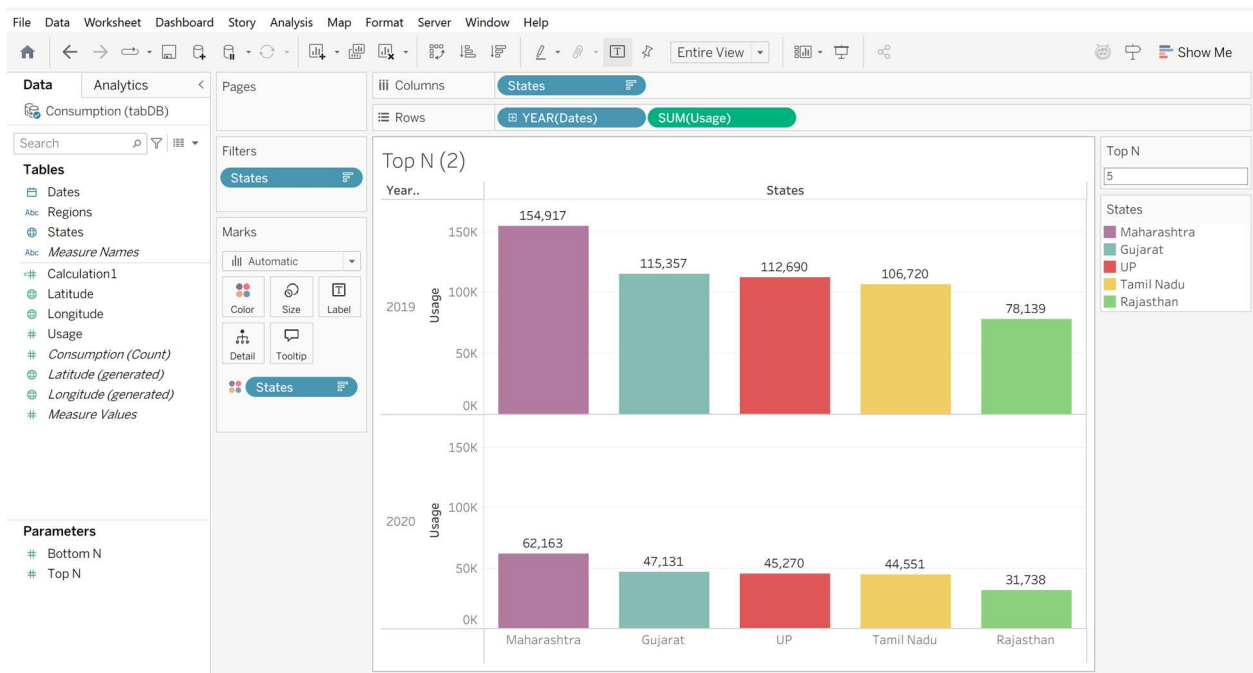
Consumption	Abc	Consumption	Consumption	Consumption	#
States	Regions	Latitude	Longitude	Dates	Consumpt
Punjab	NR	31.5200	75.9800	01/02/2019	119.9
Haryana	NR	28.4500	77.0200	01/02/2019	130.3
Rajasthan	NR	26.4500	74.6400	01/02/2019	234.1
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Chandigarh	NR	30.7200	76.7800	01/02/2019	5.0
Chhattisgarh	WR	22.0904	82.1600	01/02/2019	78.7
Gujarat	WR	22.2587	71.1924	01/02/2019	319.5
MP	WR	21.3004	76.1300	01/02/2019	253.0
Maharashtra	WR	19.2502	73.1602	01/02/2019	428.6
Goa	WR	15.4920	73.8180	01/02/2019	12.8
DNH	WR	20.2666	73.0166	01/02/2019	18.6

Data Source | Story on Electricity Consum... | 2019 State Consumption | 2020 State Consumption | Total Consumption | Usage by Region | Top N | Bottom N | 2019 Month wise Consumption

Step 3 – Data Filters

Description

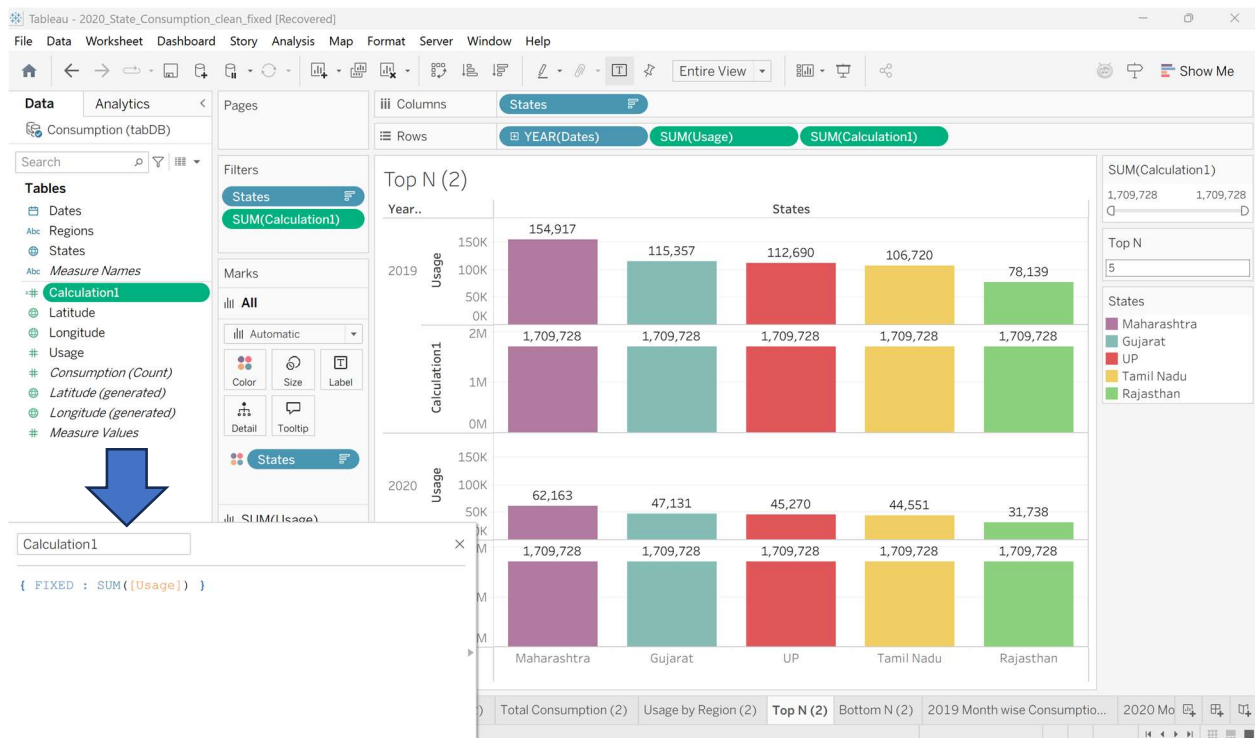
Interactive filters such as State, Region, Year, Date, Top N, and Bottom N were created to enable users to explore electricity consumption from different perspectives.



Step 4 – Calculated Fields

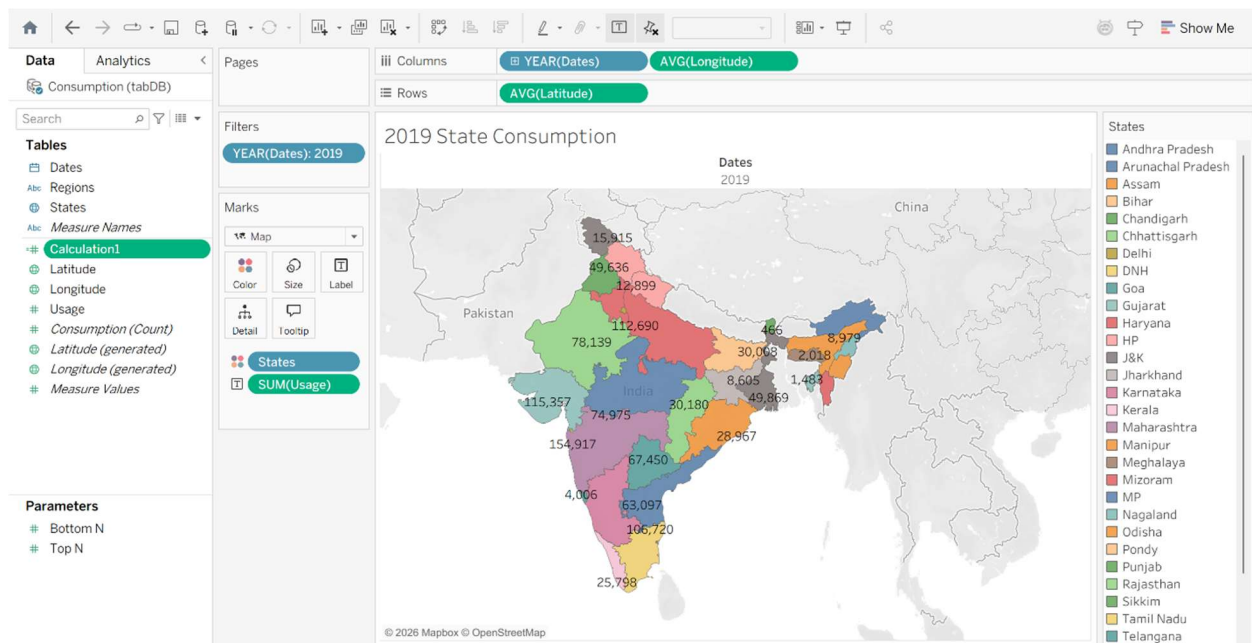
Description

Calculated fields were created to compute overall electricity consumption and perform additional analytical operations required for visualization.



Business Questions with Visualizations

1. 2019 State Consumption (Map)



Business Question

Which states recorded the highest and lowest electricity consumption in 2019?

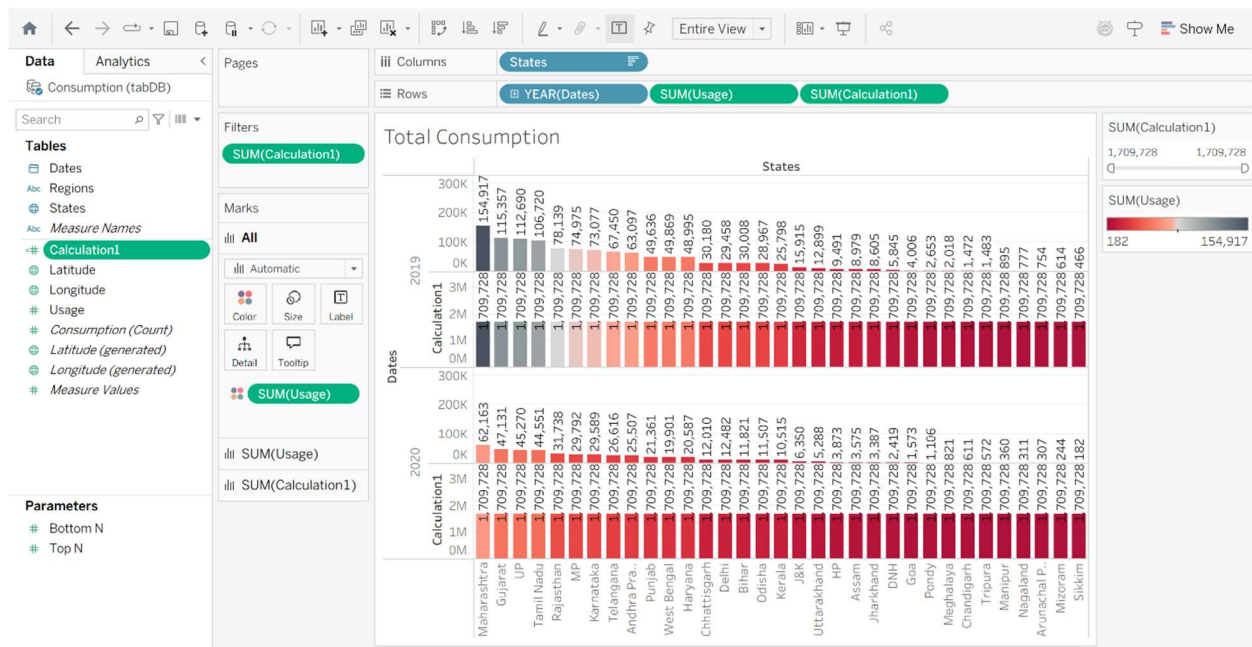
Analysis

The map shows electricity consumption across Indian states in 2019. Maharashtra recorded the highest electricity consumption (154,917 MU), followed by Gujarat (115,357 MU) and Uttar Pradesh (112,690 MU). States like Sikkim, Arunachal Pradesh and Mizoram had the lowest consumption because of smaller populations and lower industrial activity.

Business Answer

Electricity demand in 2019 was concentrated in economically developed and industrialized states, while smaller states consumed comparatively less electricity.

2. Total Consumption by State



Business Question

How did electricity consumption change across states between 2019 and 2020?

Analysis

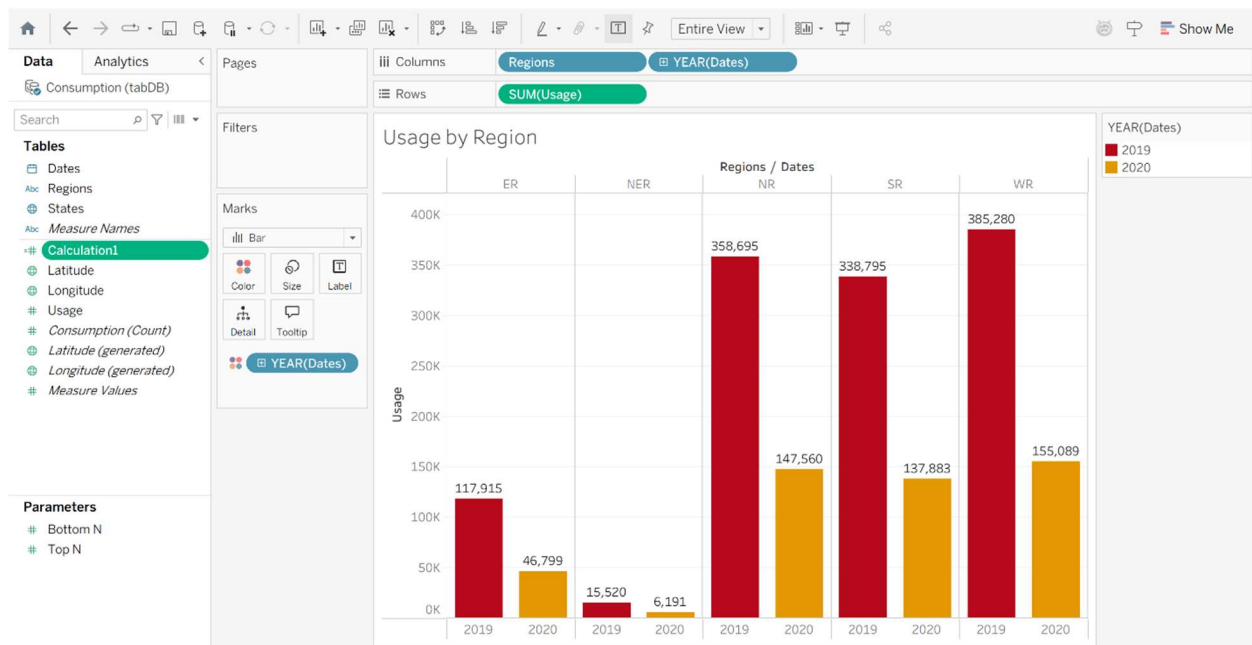
The chart compares electricity consumption of all states in 2019 and 2020. Most states experienced a noticeable decline in electricity usage during 2020.

Maharashtra remained the highest consumer in both years, although its consumption reduced significantly. Similar reductions are observed in Gujarat, Uttar Pradesh and Tamil Nadu.

Business Answer

Electricity consumption decreased across almost all states in 2020, mainly due to the COVID-19 lockdown and reduced industrial and commercial operations.

3. Usage by Region



Business Question

Which region contributed the most to India's electricity consumption?

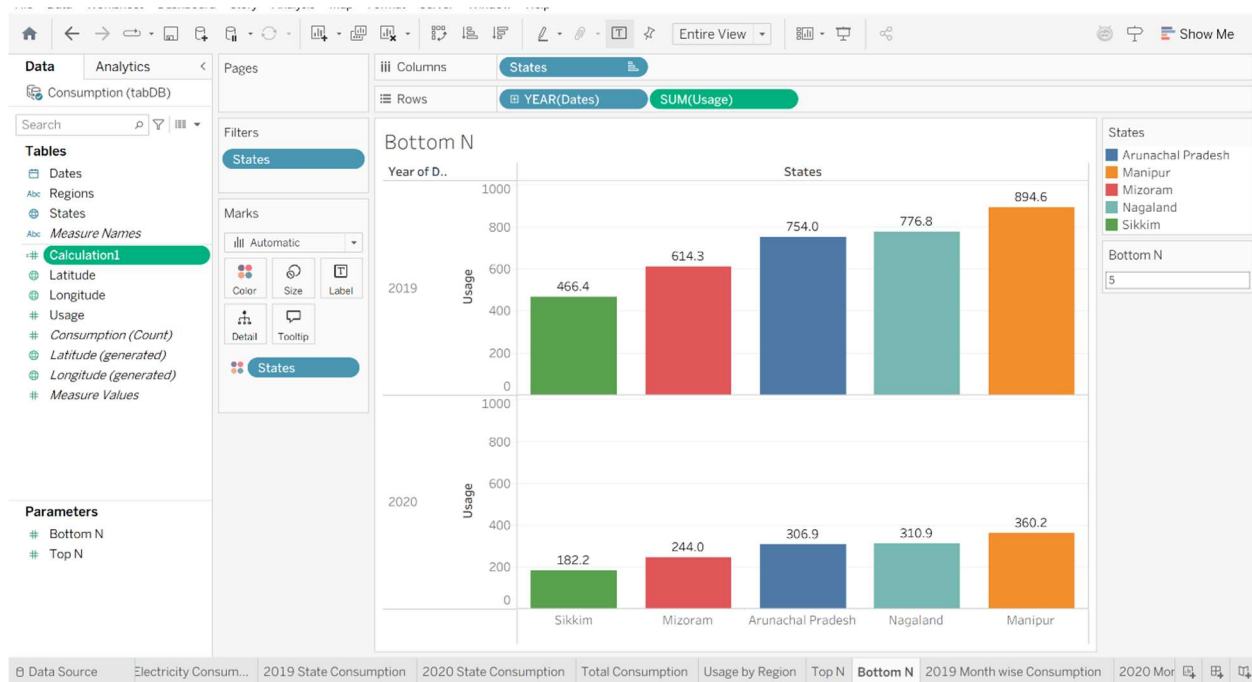
Analysis

The Western Region (WR) recorded the highest electricity consumption in both 2019 and 2020. The Northern Region (NR) and Southern Region (SR) followed closely. The North-Eastern Region (NER) contributed the smallest share. Every region experienced a decline in 2020.

Business Answer

The Western Region is the major contributor to India's electricity demand, while the North-Eastern Region has the lowest electricity consumption.

4. Bottom N States



Business Question

Which states recorded the lowest electricity consumption?

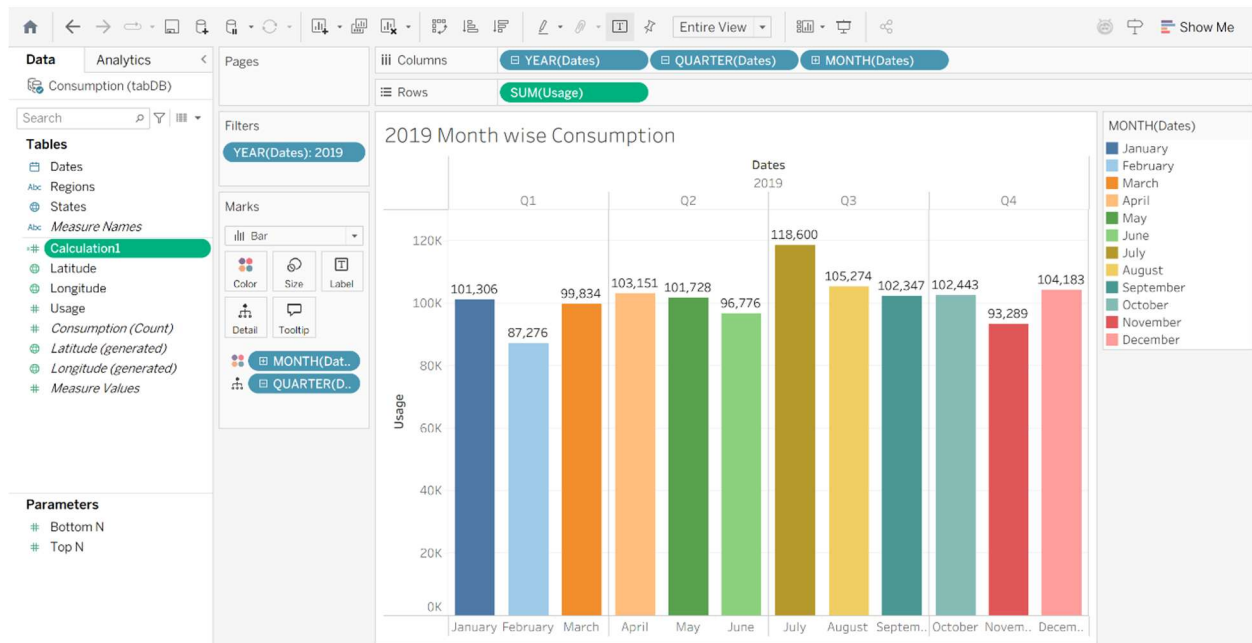
Analysis

The Bottom N visualization identifies Sikkim, Mizoram, Arunachal Pradesh, Nagaland and Manipur as the states with the lowest electricity usage in both years. These states have smaller populations, fewer industries and lower commercial electricity demand.

Business Answer

The lowest electricity consumption is consistently observed in the North-Eastern states due to lower industrialization and population.

5. 2019 Month-wise Consumption



Business Question

How did electricity consumption vary across different months in 2019?

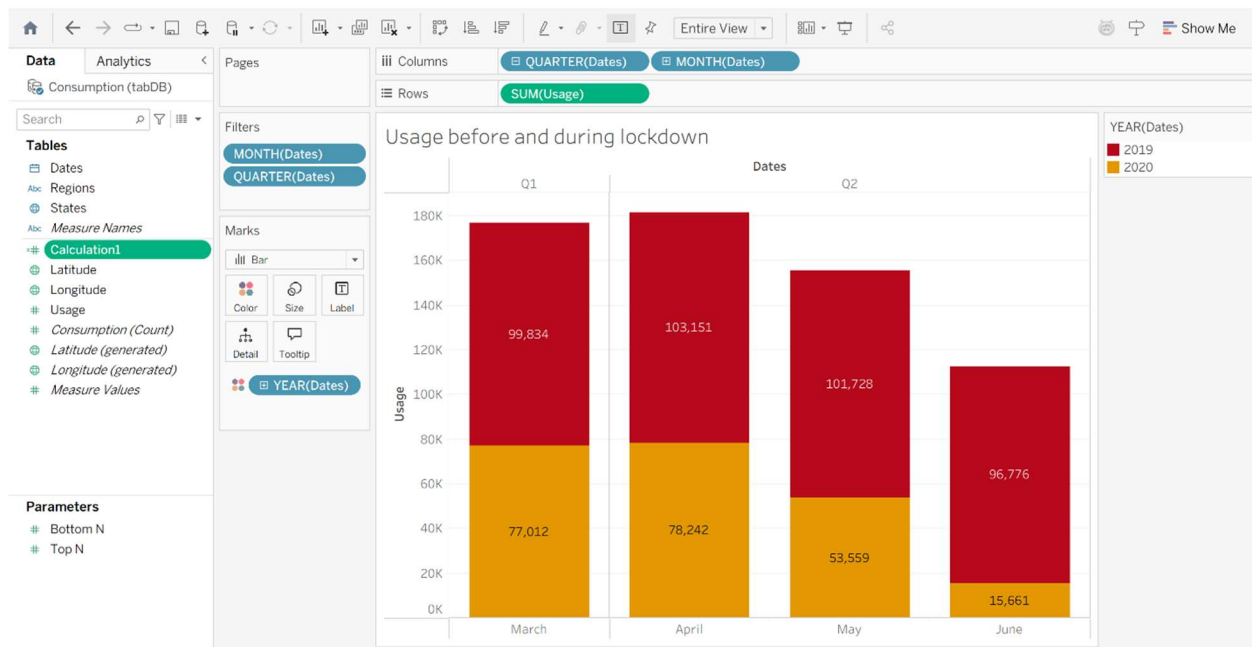
Analysis

Electricity consumption remained relatively stable throughout 2019. The highest consumption was recorded in July (118,600 MU), while February had the lowest consumption (87,276 MU). The increase during summer months indicates higher cooling and industrial electricity demand.

Business Answer

Monthly electricity consumption follows seasonal patterns, with higher usage during summer and lower usage during cooler months.

6. Usage Before and During Lockdown



Business Question

What was the impact of the COVID-19 lockdown on electricity consumption?

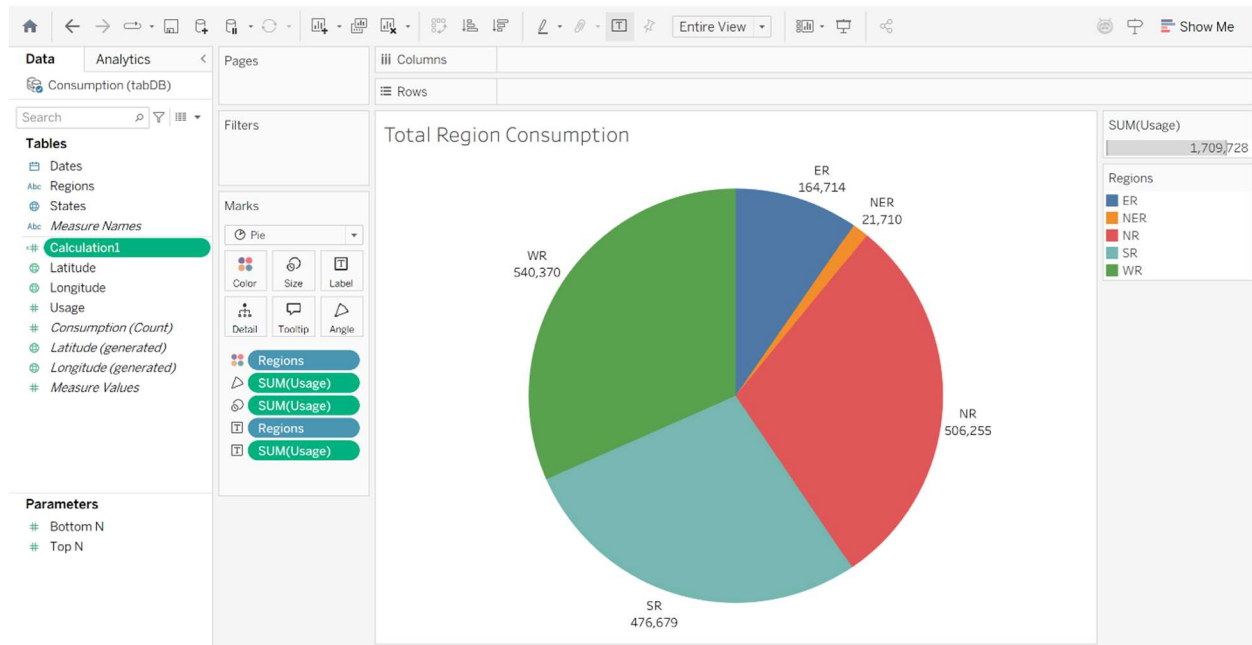
Analysis

The comparison between March and June shows that electricity consumption remained high during March and April but declined sharply during May and reached the lowest level in June 2020 (15,661 MU). This reflects the nationwide lockdown and reduced industrial activity.

Business Answer

COVID-19 significantly reduced electricity demand because industries, commercial establishments and transportation activities were temporarily shut down.

7. Total Region Consumption (Pie Chart)



Business Question

How is total electricity consumption distributed among different regions?

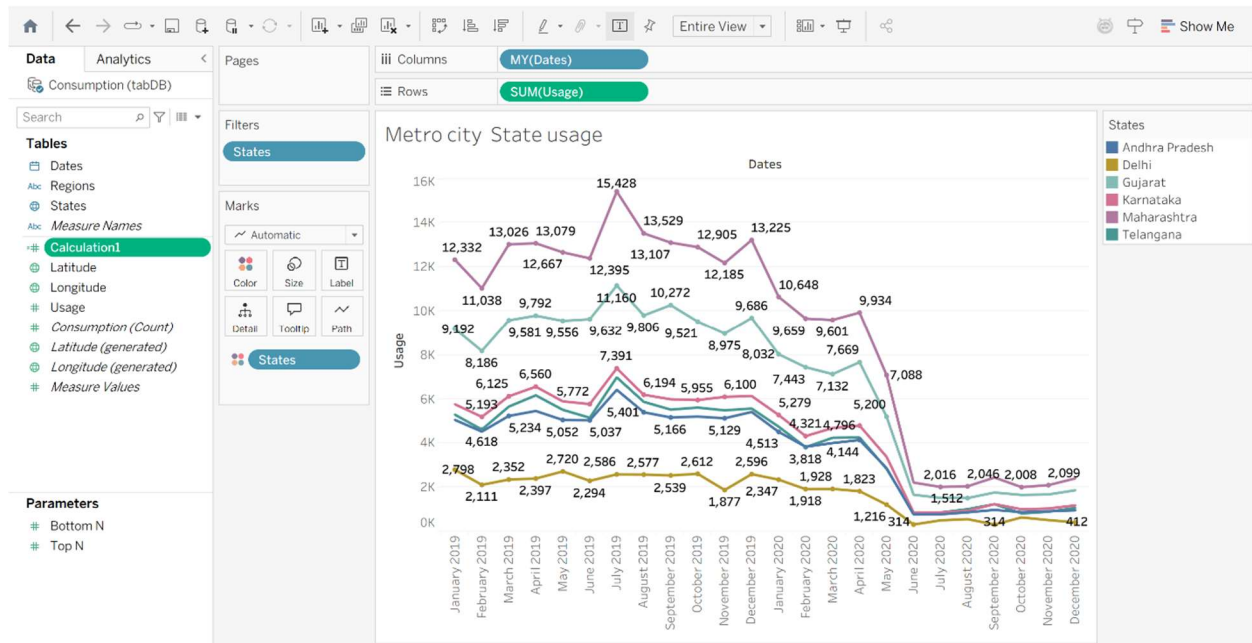
Analysis

The pie chart indicates that the Western Region contributes the highest share of electricity consumption (540,370 MU), followed by the Northern Region (506,255 MU) and Southern Region (476,679 MU). The North-Eastern Region contributes only a small fraction of the total electricity demand.

Business Answer

Electricity consumption is concentrated mainly in the Western, Northern and Southern regions due to higher industrial and economic activities.

8. Metro City State Usage (Line Chart)



Business Question

How did electricity consumption change over time among major metro city states?

Analysis

Maharashtra consistently recorded the highest electricity consumption among the selected metro states, followed by Gujarat and Karnataka. A significant decline is visible from May 2020 onwards across all states, reflecting the impact of the COVID-19 lockdown. Consumption gradually stabilized during the later months of 2020 but remained below 2019 levels.

Business Answer

Metro city states have the highest electricity demand because of urbanization and industrial growth, but they were also the most affected by the pandemic-related restrictions.
